

SECURITY OPERATIONS CENTER

INCIDENT ANALYSIS REPORT

Phishing Campaign — TryHackMe SOC Simulator

|                     |                                      |
|---------------------|--------------------------------------|
| Report Reference    | SOC-SIM-2026-001                     |
| Classification      | CONFIDENTIAL — For Training Purposes |
| Platform            | TryHackMe SOC Simulator              |
| Scenario            | Introduction to Phishing             |
| Report Date         | March 2026                           |
| Severity            | MEDIUM — Blocked at Perimeter        |
| MITRE ATT&CK Tactic | Initial Access — T1566 (Phishing)    |

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# 1. Analyst Profile

| Field                         | Details                                                                |
|-------------------------------|------------------------------------------------------------------------|
| Full Name                     | Sarmad Farooq                                                          |
| Current Role                  | MS Cybersecurity Student   Aspiring SOC Analyst / Security Engineer    |
| Academic Qualification        | MS in Cybersecurity (In Progress)                                      |
| Core Competencies             | Network Security, Threat Detection, Incident Response, SIEM Operations |
| Certifications / Foundational | CCNA (Networking Fundamentals)                                         |
| LinkedIn                      | linkedin.com/in/sarmad-farooq-cybersecurity-engineer                   |
| GitHub                        | github.com/isarmadfarooq                                               |

# 2. Professional Summary

I am a Cybersecurity graduate student currently pursuing my MS in Cybersecurity with a strong focus on Security Operations (SOC), Threat Detection, and Incident Response. With a foundation in networking (CCNA) and hands-on experience using cybersecurity labs and simulations, I aim to develop practical skills required for real-world security environments.

**This report demonstrates proficiency in the following SOC disciplines:**

- Triaging and classifying security alerts using a structured SOC methodology
- Investigating phishing indicators across email headers, domains, and embedded URLs
- Correlating alerts across SIEM, firewall, and proxy data sources
- Documenting findings in industry-standard incident report format
- Applying MITRE ATT&CK framework mappings to observed TTPs

# 3. Project Overview

| Attribute     | Value                                     |
|---------------|-------------------------------------------|
| Project Title | Introduction to Phishing — SOC Simulation |
| Platform      | TryHackMe SOC Simulator                   |

|                     |                                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| Category            | Phishing Analysis / Email Security                                                    |
| Scenario Type       | Blue Team — Security Operations Center (SOC)                                          |
| SIEM Selected       | Splunk (Commercial SIEM — chosen for out-of-the-box SOC capabilities)                 |
| Threat Category     | Phishing / Business Email Compromise (BEC)                                            |
| MITRE ATT&CK Tactic | Initial Access — T1566 (Phishing)                                                     |
| Outcome             | 3 True Positives identified; 1 False Positive correctly classified; all alerts closed |

### 3.1 Objectives

- Simulate a real-world SOC environment and respond to phishing-related alert queues
- Identify suspicious email activity using header analysis, domain verification, and URL investigation
- Analyse phishing indicators (links, domains, sender spoofing, shortened URLs)
- Classify each alert as True Positive (TP) or False Positive (FP) with documented justification
- Develop structured incident investigation and reporting skills aligned with industry standards

### 3.2 Tools & Technologies

| Tool / Technology               | Purpose                                                                        |
|---------------------------------|--------------------------------------------------------------------------------|
| Splunk SIEM (Simulated)         | Alert triage, log analysis, correlation across data sources                    |
| Email Analysis                  | Header inspection, sender domain verification, link extraction                 |
| Threat Intelligence (AbuseIPDB) | IP reputation lookup, abuse scoring, ASN / geolocation analysis                |
| Firewall / Proxy Logs           | Outbound connection monitoring, blacklist rule enforcement, block confirmation |
| MITRE ATT&CK Framework          | TTP classification and technique mapping for each threat actor behaviour       |
| SOC Investigation Workflow      | Alert assignment, case creation, triage, classification, and formal closure    |

## 4. Executive Summary

This report documents the investigation of four security alerts generated within the TryHackMe SOC Simulator environment. All alerts were triggered by inbound phishing emails or correlated firewall events targeting employees of the simulated organisation, thetrydaily.thm.

Of the four alerts investigated, three were classified as True Positives involving credential-harvesting phishing campaigns impersonating Amazon and Microsoft. One alert was correctly classified as a False Positive following SIEM correlation that confirmed the sender as a legitimate third-party HR partner.

A critical attack sequence was identified between Alerts 8815 and 8816, where a phishing email successfully induced a user click-through on a shortened malicious URL (bit.ly/3sHkX3da12340), which was subsequently blocked by the organisation's firewall perimeter control — confirming both alerts as correlated True Positives.

## 4.1 Alert Summary Matrix

| Alert ID | Title                                                      | Classification | Disposition                           |
|----------|------------------------------------------------------------|----------------|---------------------------------------|
| 8814     | Inbound Email — Suspicious External Link                   | False Positive | Closed — No Action Required           |
| 8815     | Inbound Email — Suspicious External Link (Amazon Spoof)    | True Positive  | Closed — Blocked by FW, No Escalation |
| 8816     | Access to Blacklisted External URL Blocked by Firewall     | True Positive  | Closed — Correlated with Alert 8815   |
| 8817     | Inbound Email — Suspicious External Link (Microsoft Spoof) | True Positive  | Closed — Contained, No Click Detected |

## 4.2 Key Findings

1. Three phishing campaigns identified; all employing brand impersonation (Amazon, Microsoft) and social engineering tactics (urgency, account security, parcel delivery).
2. A URL-click event was observed and confirmed blocked at the perimeter firewall (Alert 8816), correlating to the phishing email in Alert 8815 — confirming successful threat containment.
3. The typosquatted domain m1crosoftsupport.co (replacing "i" with "1") demonstrates deliberate obfuscation to evade basic domain validation.
4. Alert 8814 was correctly identified as a False Positive following SIEM query that confirmed hrconnex.thm as a legitimate onboarding partner communicated to staff via prior internal email.
5. No lateral movement or post-exploitation activity was observed across any alert — the attack chain was contained at the Initial Access stage.

# 5. Scenario Walkthrough

The following section provides a step-by-step walkthrough of the SOC Simulator scenario, documenting the analyst workflow from initial dashboard access through to alert closure.

## 5.1 Starting the Scenario — SIEM Selection

Upon launching the TryHackMe SOC Simulator, the analyst is prompted to select a SIEM platform for the simulation environment. Two primary options are presented:

| Platform  | Splunk                               | Elastic Stack / Sentinel                 |
|-----------|--------------------------------------|------------------------------------------|
| Type      | Commercial, Proprietary              | Open-Source / Cloud-Native               |
| Use Case  | Enterprise SOC — Out-of-the-box SIEM | Customisable, cost-effective deployments |
| Focus     | Ease of use, powerful search (SPL)   | Microsoft ecosystem (Sentinel)           |
| Selected? | YES — Selected for this simulation   | No                                       |

Splunk was selected due to its widespread adoption across enterprise SOC environments and its powerful Search Processing Language (SPL) for log correlation.

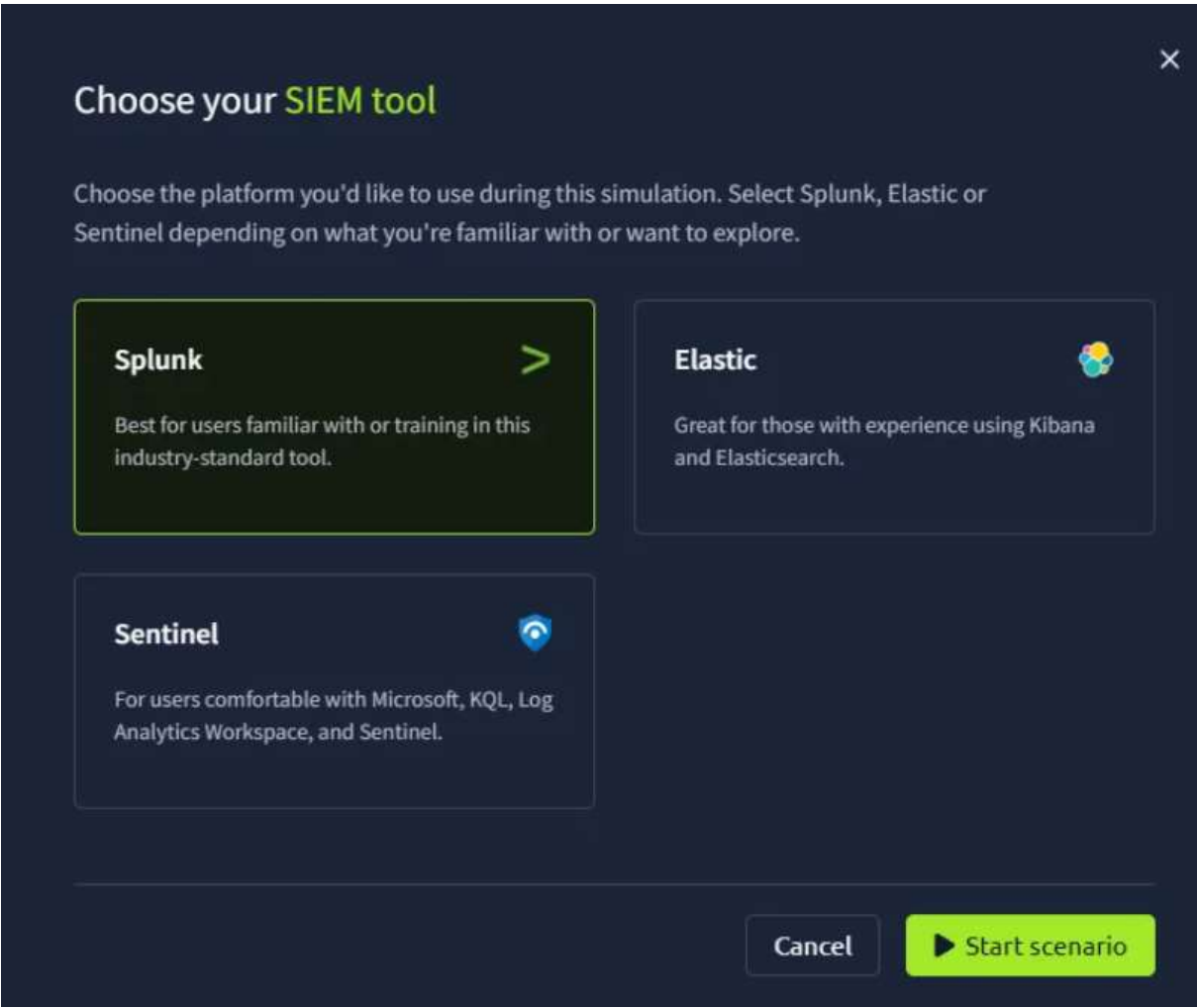


Figure 1 — SIEM Platform Selection Screen (TryHackMe SOC Simulator)

### 5.2 Alert Queue — Dashboard Overview

Following scenario initialisation, the analyst is presented with the SOC dashboard. All active alerts are accessible via the Alert Queue, accessible through the left-side navigation menu. The queue displays unassigned alerts sorted by arrival timestamp and severity.

Industry Best Practice: In a production SOC environment, analysts should always prioritise alerts by severity (Critical → High → Medium → Low) before investigating in order of arrival. For the purposes of this walkthrough, alerts are documented in arrival order for clarity.

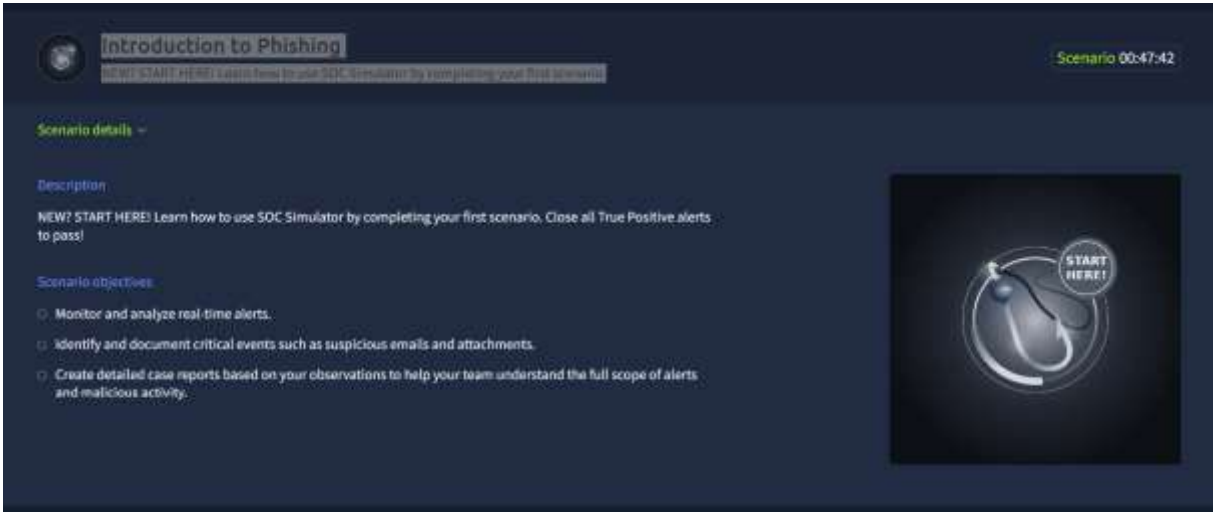


Figure 2 — SOC Dashboard Alert Queue Overview

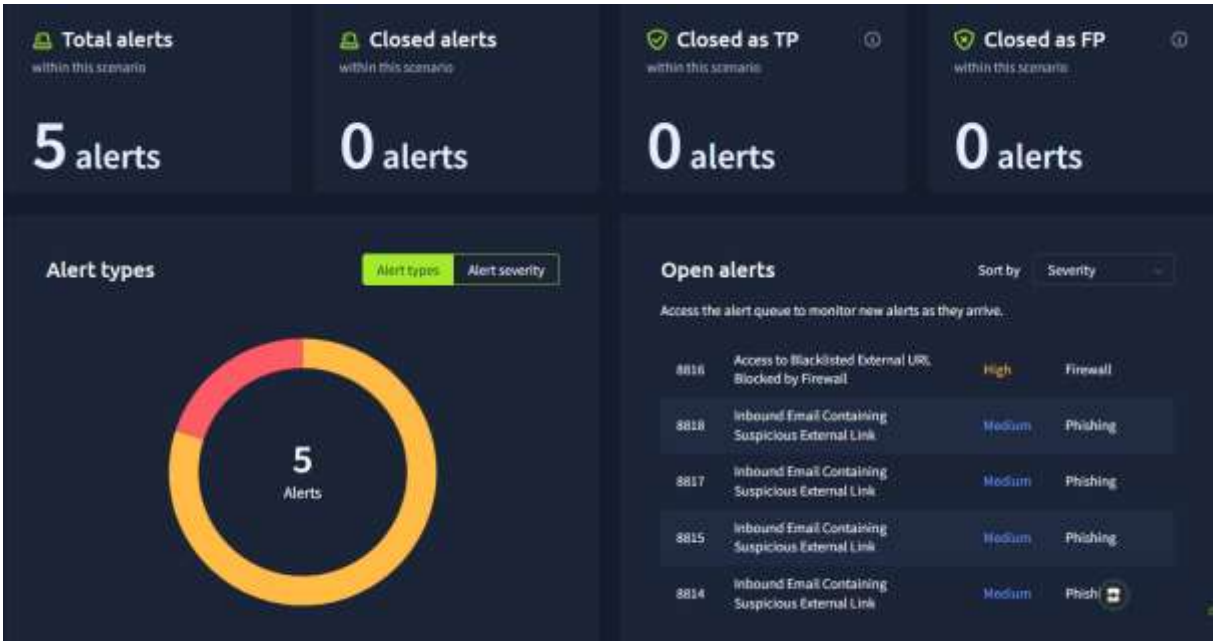


Figure 3 — Alert Queue Expanded View Showing All Active Alerts

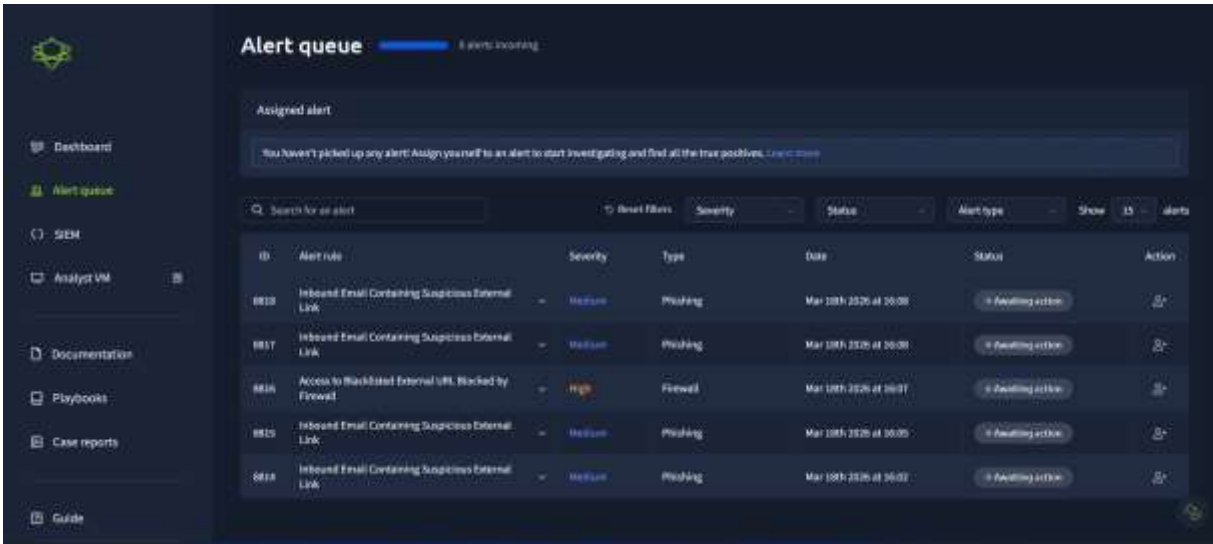


Figure 4 — Alert Queue Navigation via Left-Side Menu Panel

## 6. Alert Investigations

### 6.1 Alert 8814 — Inbound Email Containing Suspicious External Link

| Alert Attribute      | Value                                             |
|----------------------|---------------------------------------------------|
| Alert ID             | 8814                                              |
| Title                | Inbound Email Containing Suspicious External Link |
| Data Source          | Email                                             |
| Timestamp            | 03/18/2026 16:00:46.213                           |
| Subject              | Action Required: Finalize Your Onboarding Profile |
| Sender               | onboarding@hrconnex.thm                           |
| Recipient            | j.garcia@thetrydaily.thm                          |
| Attachment           | None                                              |
| Direction            | Inbound                                           |
| Final Classification | FALSE POSITIVE                                    |

#### 6.1.1 Alert Description

This alert was triggered by an inbound email containing one or more external links exhibiting potentially suspicious characteristics. Per SOC standard procedure, the analyst must check firewall and proxy logs to determine whether any endpoint attempted to access the URLs embedded in the email and whether those connections were permitted or blocked.



6.1.2 Email Content Analysis

The email body reads as an HR onboarding notification directed to a new employee (j.garcia), requesting completion of an onboarding profile setup via a hyperlink. The embedded URL points to: <https://hrconnex.thm/onboarding/15400654060/j.garcia>

Initial Assessment — Indicators to Investigate:

- External domain link embedded in email (hrconnex.thm) — warrants domain reputation check
- Urgency language: "Action Required" — common social engineering trigger
- Personalised URL containing the recipient's username — potential targeting indicator
- Sent from external domain rather than internal HR department

6.1.3 SIEM Correlation & Investigation

The analyst queried the SIEM for all log entries related to the domain hrconnex.thm. The investigation returned a critical data point: a prior internal email from the HR department had been sent to all staff prior to this alert, explicitly notifying employees that onboarding communications would arrive from the third-party partner domain hrconnex.thm.

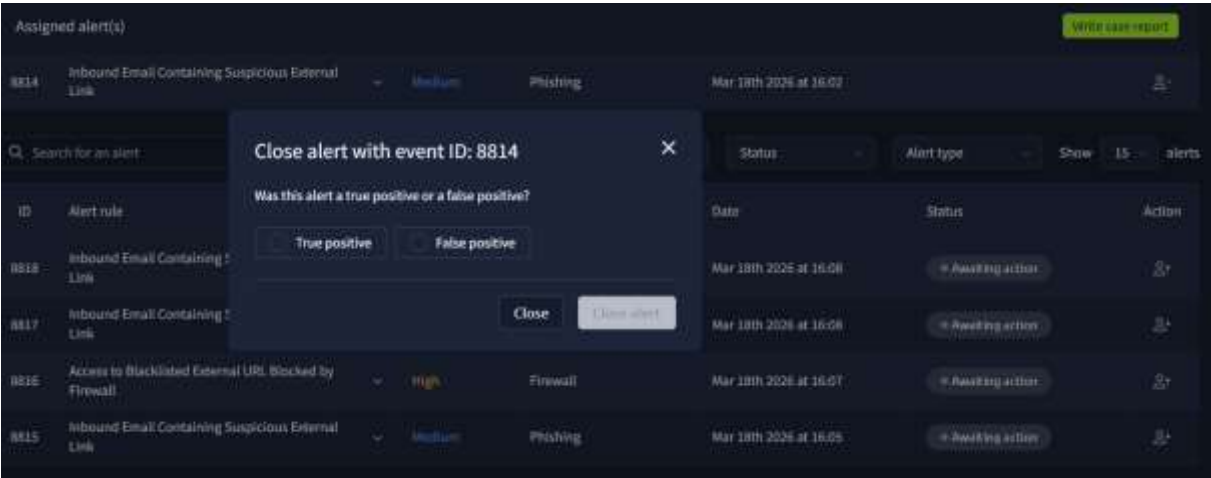
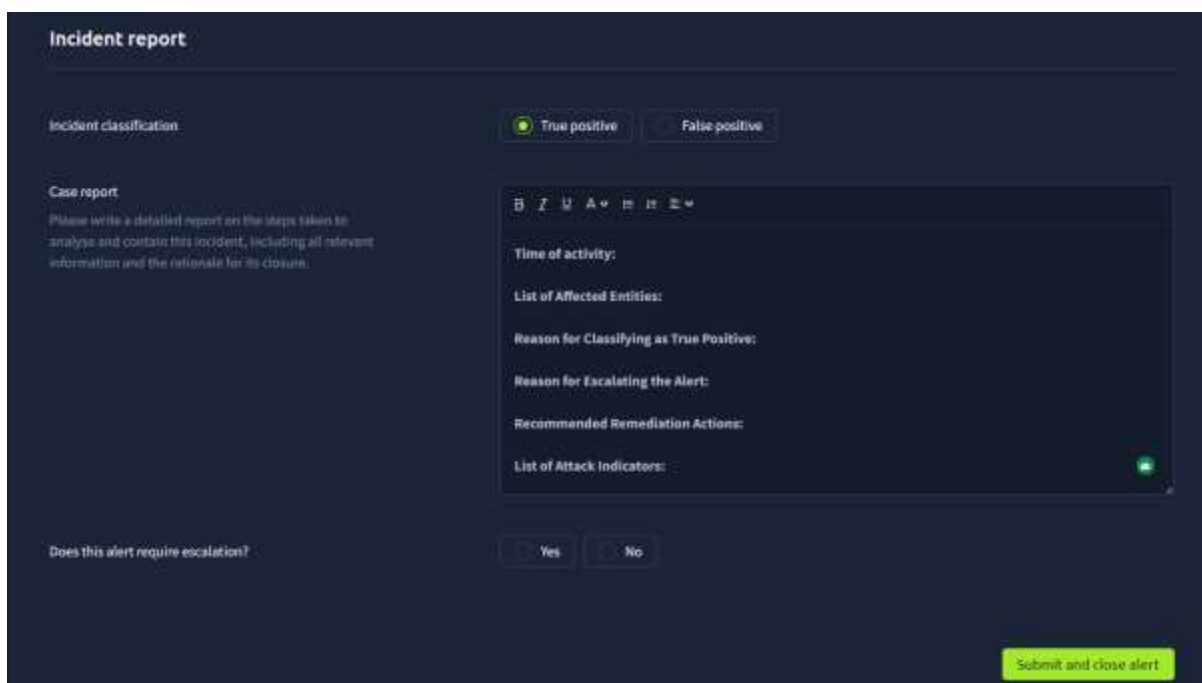


Figure 5 — Alert 8814 Assignment and Initial Triage Screen



The interface is titled "Incident report" and contains several sections for creating a case report. At the top, there is an "Incident classification" section with two radio buttons: "True positive" (selected) and "False positive". Below this is a "Case report" section with a text area for writing a detailed report. To the right of the text area is a list of fields to be filled out: "Time of activity:", "List of Affected Entities:", "Reason for Classifying as True Positive:", "Reason for Escalating the Alert:", "Recommended Remediation Actions:", and "List of Attack Indicators:". At the bottom left, there is a "Does this alert require escalation?" section with "Yes" and "No" radio buttons. A green "Submit and close alert" button is located at the bottom right.

Figure 6 — Incident Case Report Creation Interface

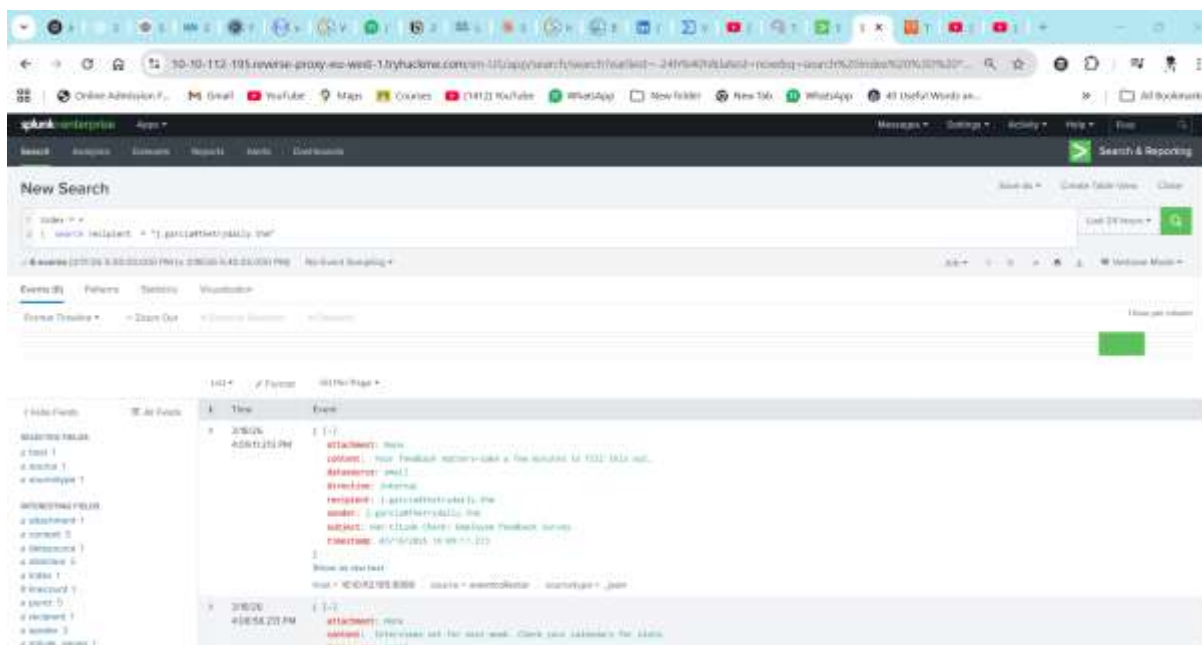


Figure 7 — SIEM Query Results: Prior Internal HR Email Confirming hrconnex.thm as Legitimate Partner



Figure 8 — SIEM Log Detail Confirming Legitimate Sender Identity

### 6.1.4 Investigation Findings

The SIEM correlation exercise produced definitive evidence that contradicted the initial suspicion raised by the automated alert:

- The domain hrconnex.thm is confirmed as a legitimate third-party HR onboarding partner
- The organisation's HR department pre-notified staff of communications from this domain
- No malicious URL activity was detected in firewall or proxy logs for this domain
- The personalised URL structure (/j.garcia) is consistent with legitimate onboarding workflows

This alert represents a common benign trigger in email security monitoring — automated rules flagging any external link regardless of domain reputation. The SIEM correlation step is essential to avoid false positive alert fatigue.

### 6.1.5 Classification & Disposition

| Classification Field | Value                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Verdict              | <b>FALSE POSITIVE</b>                                                                               |
| Justification        | SIEM query confirmed hrconnex.thm as pre-approved HR partner domain; no malicious activity detected |
| Response Action      | Alert closed — no further action required; case documented                                          |
| Escalation Required  | No                                                                                                  |

The screenshot shows a web-based incident report form. At the top, it says 'Incident report'. Below that, there's a section for 'Incident classification' with two radio buttons: 'True positive' and 'False positive'. The 'False positive' button is selected. Underneath, there's a 'Closure rationale' section with a text area for explaining why the incident is a false positive. To the right of the text area, there's a summary box containing the following information: 'Time of Activity:', 'List of Related Entities:' with 'User origin: onboarding@hrconnex.thm' and 'User impacted: j.garcia@thetrydaily.thm', and 'Reason for Classifying as False Positive:' with the text 'Everything is normally associated with an onboarding process, nothing malicious. The link 'https://hrconnex.thm/onboarding/15400654060/j.garcia' that has triggered the alert is not showing any result.' At the bottom right, there is a green button labeled 'Submit and close alert'.

Figure 9 — Incident Report: Alert 8814 Classified as False Positive

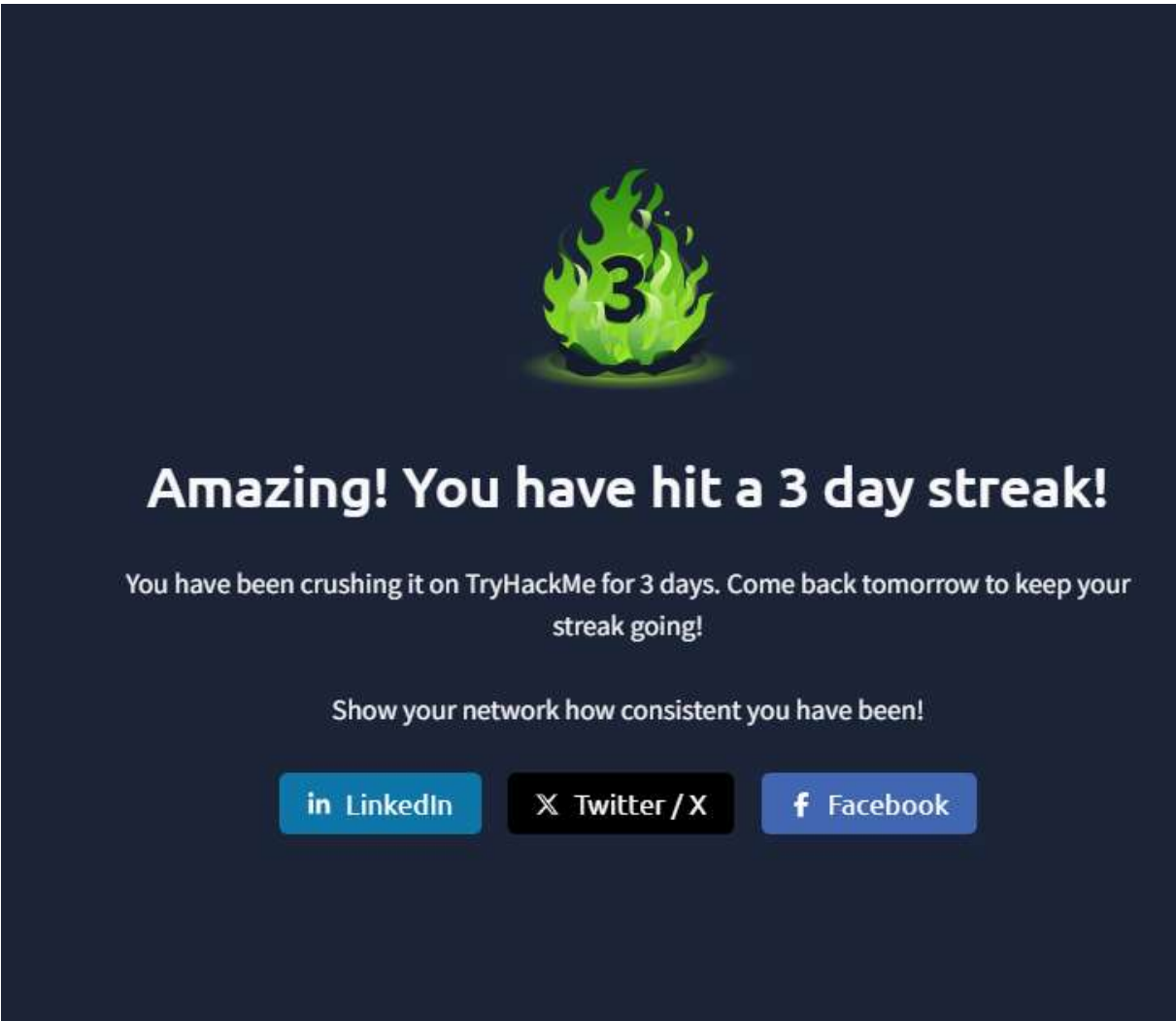


Figure 10 — Case Closure Confirmation for Alert 8814



Figure 11 — Alert 8814 Closed Status Indicator

6.2 Alert 8815 — Inbound Email Containing Suspicious External Link (Amazon Brand Impersonation)

| Alert Attribute | Value                                                       |
|-----------------|-------------------------------------------------------------|
| Alert ID        | 8815                                                        |
| Title           | Inbound Email Containing Suspicious External Link           |
| Data Source     | Email / Network Security                                    |
| Timestamp       | 11/20/2025 20:03:09.489                                     |
| Subject         | Your Amazon Package Couldn't Be Delivered — Action Required |
| Sender          | urgents@amazon.biz                                          |

|                      |                          |
|----------------------|--------------------------|
| Recipient            | h.harris@thetrydaily.thm |
| Attachment           | None                     |
| Direction            | Inbound                  |
| Final Classification | TRUE POSITIVE            |

### 6.2.1 Alert Description

This alert was triggered by an inbound email containing one or more external links flagged for suspicious characteristics. The investigation requires firewall and proxy log review to determine whether endpoint click-through occurred and whether the outbound connection was permitted or blocked.

### 6.2.2 Email Content Analysis

The email impersonates Amazon's delivery notification service, claiming a parcel could not be delivered due to an incomplete address. The recipient is directed to click a bit.ly shortened URL (<http://bit.ly/3sHkX3da12340>) to "confirm shipping information" within 48 hours — creating a false sense of urgency.

#### Phishing Indicators Identified:

- Sender domain amazon.biz — Amazon's legitimate domain is amazon.com. The .biz TLD is a clear impersonation indicator
- URL shortening via bit.ly — obscures the true destination, a common phishing obfuscation technique
- Urgency language: "48 hours... returned to sender" — classic social engineering pressure tactic
- Generic salutation "Dear Customer" — inconsistent with legitimate Amazon communications which include the account holder's name
- No package tracking number referenced — inconsistent with genuine delivery notifications

### 6.2.3 SIEM Correlation & Investigation

The analyst queried the SIEM for the bit.ly URL (<http://bit.ly/3sHkX3da12340>) and the sender domain amazon.biz. The SIEM results confirmed that recipient h.harris had attempted to access the embedded URL. Cross-referencing with firewall logs revealed the outbound connection to 67.199.248.11:80 was blocked by the organisation's perimeter firewall — correlating directly to Alert 8816.

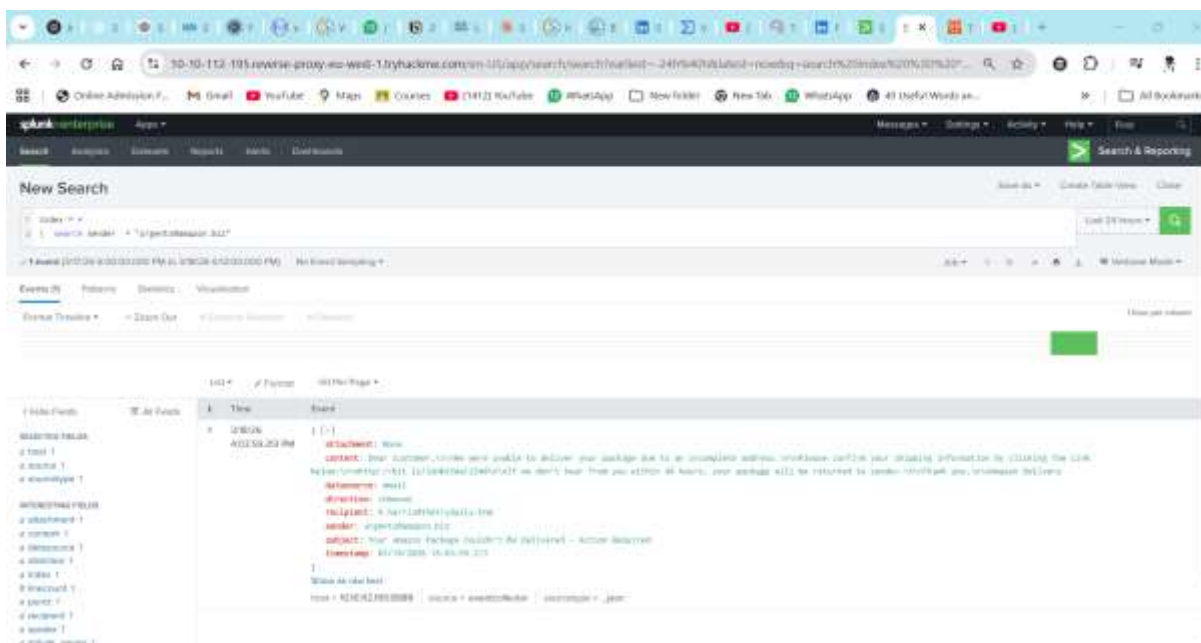


Figure 12 — SIEM Query Results: URL Click Attempt Detected for Alert 8815

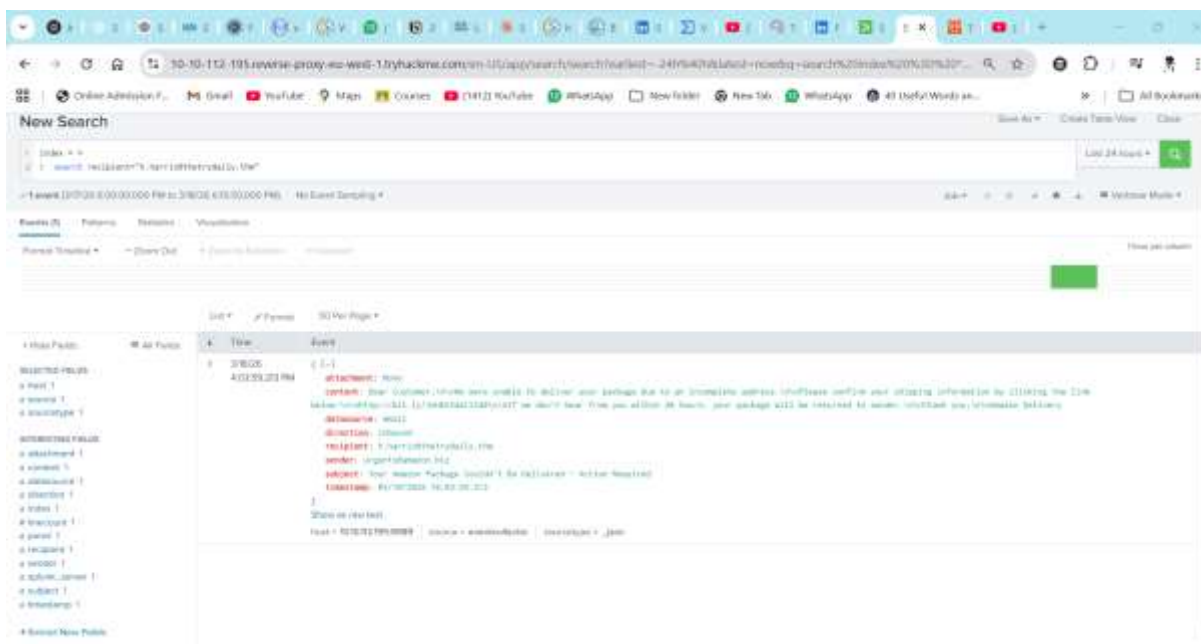


Figure 13 — SIEM Log: Firewall Block Event Correlated to Bit.ly URL

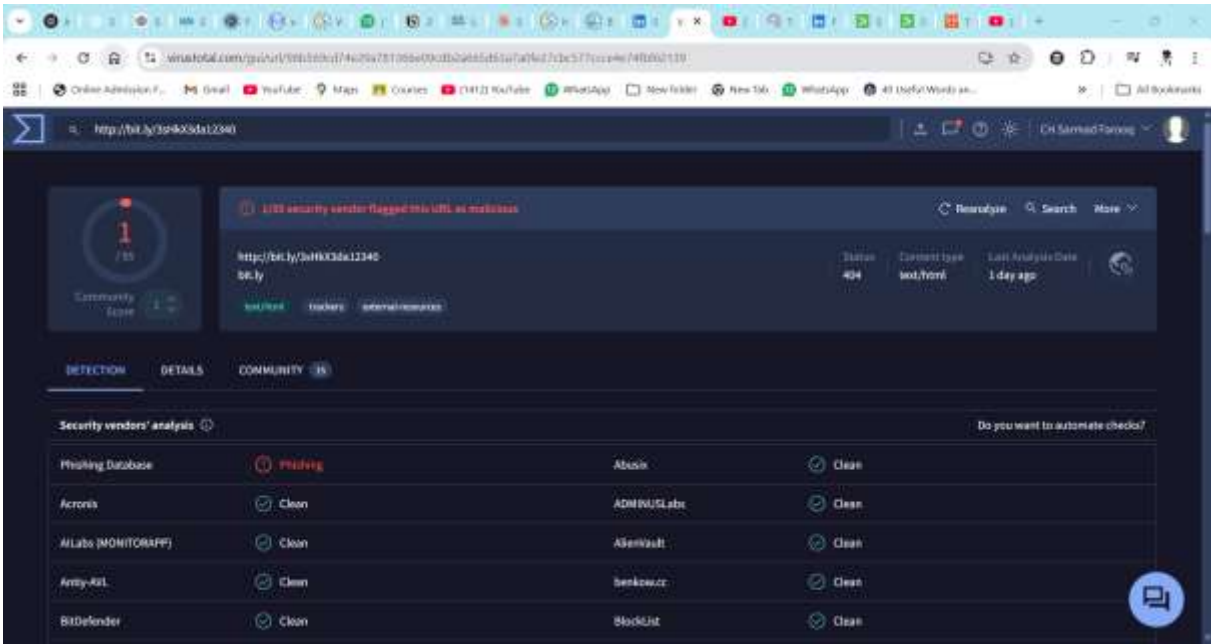


Figure 14 — Expanded SIEM View: Full Event Timeline for Alert 8815

6.2.4 Classification & Disposition

| Classification Field | Value                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Verdict              | TRUE POSITIVE                                                                                                            |
| Justification        | Spoofed sender domain (amazon.biz), bit.ly redirect URL, click-through confirmed by SIEM, connection blocked by firewall |
| Response Action      | Alert closed — firewall block confirmed; no escalation required as connection was prevented                              |
| Escalation Required  | No — threat contained at perimeter; user awareness communication recommended                                             |
| Correlated Alert     | Alert 8816 (Firewall Block — same URL)                                                                                   |

Incident classification

True positive

False positive

Case report

Please write a detailed report on the steps taken to analyse and contain this incident, including all relevant information and the rationale for its closure.

Time of activity: 3/18/26 4:03:59.213 PM

List of Affected Entities:  
Origin User: urgents@amazon.biz  
User impacted: h.harris@thetrydaily.thm

Reason for Classifying as True Positive:  
The alert has been triggered when the user 'urgents@amazon.biz' has tried to send a phishing link to the recipient 'h.harris@thetrydaily.thm'. The sender was trying to mimic Amazon domain, but 'amazon.biz' is not associated with Amazon. Additionally the link was redirecting the user to an external domain not associated with amazon, most probably a phishing attempt to harvest user credentials.

A 24-hr on the users involved just show this event.

Reason for Escalating the Alert:

Recommended Remediation Actions:

Does this alert require escalation?

Yes

No

Figure 15 — Incident Case Report: Alert 8815 True Positive Classification

Incident classification

True positive

False positive

Case report

Please write a detailed report on the steps taken to analyse and contain this incident, including all relevant information and the rationale for its closure.

recipient 'h.harris@thetrydaily.thm'. The sender was trying to mimic Amazon domain, but 'amazon.biz' is not associated with Amazon. Additionally the link was redirecting the user to an external domain not associated with amazon, most probably a phishing attempt to harvest user credentials.

A 24-hr on the users involved just show this event.

Reason for Escalating the Alert:

Recommended Remediation Actions:  
Enable or tune the email filter and the spam detection.  
Implement or configure the EDR.  
Quarantine the email with suspicious domain.  
Force password reset for the impacted user.

List of Attack Indicators:  
- Malicious Link: http://bit.ly/3iH4X3da12340  
- Not a real amazon domain.

Does this alert require escalation?

Yes

No

Figure 16 — Case Closure Documentation for Alert 8815

8815

Inbound Email Containing Suspicious External Link

Medium

Phishing

Mar 18th 2026 at 16:05

Closed

Figure 17 — Alert 8815 Closed Status Indicator

6.3 Alert 8816 — Access to Blacklisted External URL Blocked by Firewall

| Alert Attribute | Value |
|-----------------|-------|
| Alert ID        | 8816  |

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|                      |                                                        |
|----------------------|--------------------------------------------------------|
| Title                | Access to Blacklisted External URL Blocked by Firewall |
| Data Source          | Firewall / Network Security                            |
| Timestamp            | 11/20/2025 20:04:23.489                                |
| Firewall Action      | <b>BLOCKED</b>                                         |
| Source IP            | 10.20.2.17 (Internal Host — h.harris endpoint)         |
| Source Port          | 34257                                                  |
| Destination IP       | 67.199.248.11                                          |
| Destination Port     | 80 (HTTP)                                              |
| URL                  | http://bit.ly/3sHkX3da12340                            |
| Application          | web-browsing                                           |
| Protocol             | TCP                                                    |
| Firewall Rule        | Blocked Websites                                       |
| Final Classification | <b>TRUE POSITIVE</b>                                   |

### 6.3.1 Alert Description

This alert was generated when a user attempted to access an external URL present on the organisation's blacklist or threat intelligence feed. The firewall successfully blocked the outbound TCP connection on Port 80, preventing the user from reaching the malicious destination. The blacklist covers known threats, but does not guarantee protection against zero-day or newly registered malicious domains.

### 6.3.2 Technical Analysis

**Alert Correlation:** This firewall alert was generated exactly 74 seconds after the email in Alert 8815 was timestamped, confirming the chronological attack chain: email delivery → user click → firewall block.

The destination IP 67.199.248.11 is a known bit.ly infrastructure address. The URL <http://bit.ly/3sHkX3da12340> was the same shortened URL embedded in the Alert 8815 phishing email. The firewall's Blocked Websites rule matched the domain against the organisation's threat intelligence feed, blocking the connection before any data exchange occurred.

#### Network Forensics Summary:

- Source IP 10.20.2.17 belongs to h.harris — the recipient of the phishing email in Alert 8815
- HTTP (Port 80) rather than HTTPS (Port 443) indicates the phishing redirect may not use TLS — a further indicator of malicious intent
- TCP connection initiated but blocked at perimeter — no data exfiltration or credential submission occurred
- 74-second gap between email delivery and click-through indicates the user read and acted on the email immediately

The screenshot shows the Splunk web interface. At the top, there's a navigation bar with 'Search', 'Analytics', 'Dashboards', 'Reports', 'Alerts', and 'Dashboards'. A 'New Search' bar is visible with the query: `index=firewall sourcetype=* "8816, 2, 1"`. Below the search bar, there's a table of search results. The first result is highlighted, showing details for a blocked connection.

| Time                 | Event                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4/15/2024 4:15:20 PM | <pre>{   "action": "Blocked",   "application": "web-browser",   "destinationip": "10.10.10.1",   "destinationport": 80,   "protocol": "TCP",   "rule": "Blocked WebSite",   "sourceip": "10.10.10.1",   "sourceport": 54321,   "uid": "10000000000000000000",   "username": "Firewall",   "timestamp": "2024-04-15T16:15:20.000Z" }</pre> |

Figure 18 — Firewall Log: Alert 8816 Connection Block Detail

The screenshot shows the Splunk web interface. At the top, there's a navigation bar with 'Search', 'Analytics', 'Dashboards', 'Reports', 'Alerts', and 'Dashboards'. A 'New Search' bar is visible with the query: `index=firewall sourcetype=* "8816, 2, 1"`. Below the search bar, there's a table of search results. The first result is highlighted, showing details for a blocked connection. An 'Action' dialog box is open, showing the details of the blocked connection.

| Time                 | Event                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4/15/2024 4:15:20 PM | <pre>{   "action": "Blocked",   "application": "web-browser",   "destinationip": "10.10.10.1",   "destinationport": 80,   "protocol": "TCP",   "rule": "Blocked WebSite",   "sourceip": "10.10.10.1",   "sourceport": 54321,   "uid": "10000000000000000000",   "username": "Firewall",   "timestamp": "2024-04-15T16:15:20.000Z" }</pre> |

Figure 19 — Firewall Log: Destination IP and Port Information

The screenshot shows the Splunk web interface. At the top, there's a navigation bar with 'Search', 'Analytics', 'Dashboards', 'Reports', 'Alerts', and 'Dashboards'. A 'New Search' bar is visible with the query: `index=firewall sourcetype=* "8816, 2, 1"`. Below the search bar, there's a table of search results. The first result is highlighted, showing details for a blocked connection.

| Time                 | Event                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4/15/2024 4:15:20 PM | <pre>{   "action": "Blocked",   "application": "web-browser",   "destinationip": "10.10.10.1",   "destinationport": 80,   "protocol": "TCP",   "rule": "Blocked WebSite",   "sourceip": "10.10.10.1",   "sourceport": 54321,   "uid": "10000000000000000000",   "username": "Firewall",   "timestamp": "2024-04-15T16:15:20.000Z" }</pre> |

Figure 20 — Firewall Alert 8816 Full Event Details

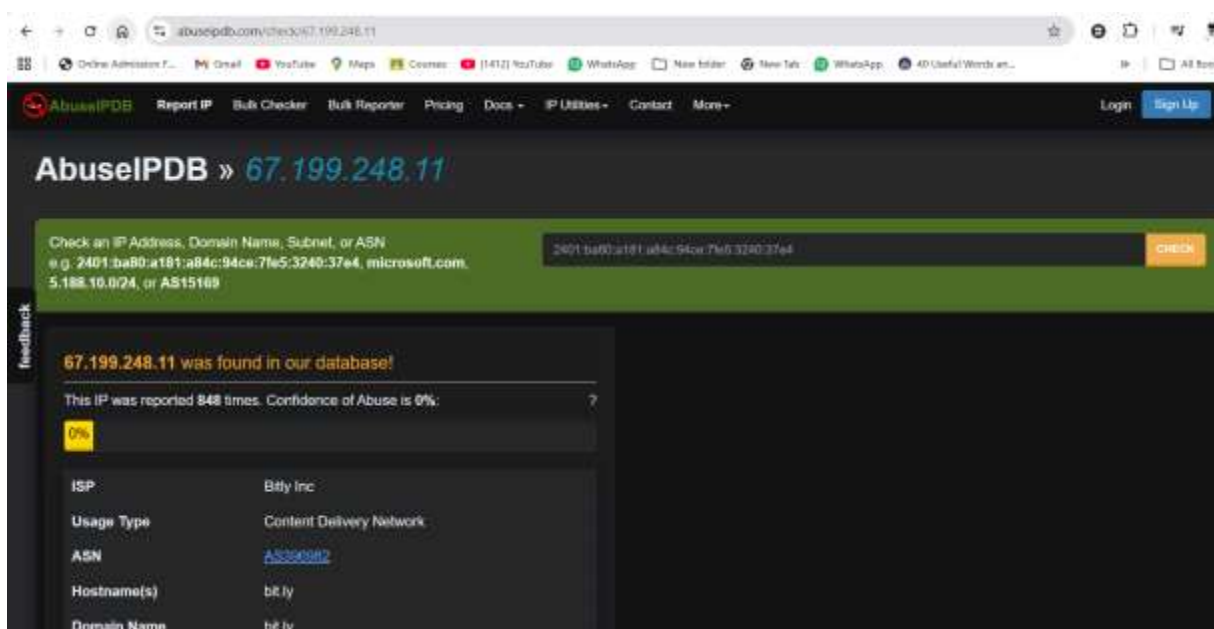


Figure 21 — SIEM Cross-Correlation View: Alert 8816 Linked to 8815

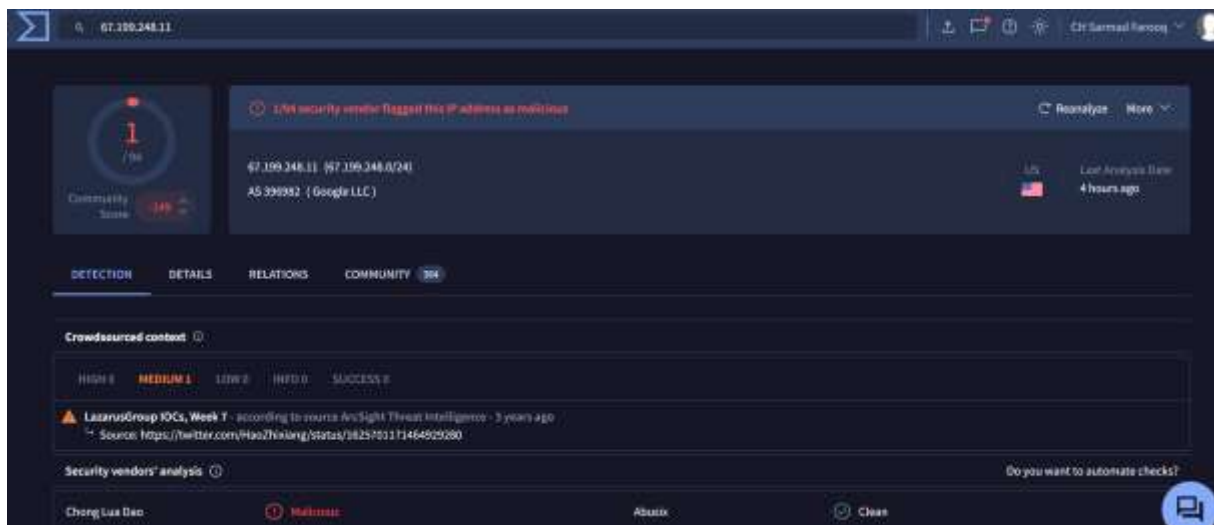


Figure 22 — Incident Report: Alert 8816 Classification

## Incident report

Incident classification

True positive

False positive

Case report

Please write a detailed report on the steps taken to analyse and contain this incident, including all relevant information and the rationale for its closure.

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Time of activity:

03/18/2025 15:05:13.111

List of Affected Entities:

- SourceIP: 10.25.2.17
- DestinationIP: 67.199.248.11

Reason for Classifying as True Positive:

The user 'h.harms@thetrdaily.shm' has received phishing email originated by the user 'urgenta@amazon.br' with a malicious link. That alert has been triggered because the user has clicked the link, the access has been blocked by the firewalls.

Reason for Escalating the Alert:

Possibility of user been compromised.

Recommended Remediation Actions:

- Force the user to reset the password.
- Enforce use of MFA.

Figure 23 — Case Documentation for Alert 8816

• Implement an MFA.

List of Attack Indicators:

• Malicious URL: <http://bit.ly/3sHwX3ds12340/>

Does this alert require escalation?

☒ Yes

☐ No

Submit and close alert

Figure 24 — Final Case Report Summary for Alert 8816

| Alert ID | Alert Name                                             | Severity | Category | Timestamp              | Status | Actions                                                                               |
|----------|--------------------------------------------------------|----------|----------|------------------------|--------|---------------------------------------------------------------------------------------|
| 8816     | Access to Blacklisted External URL Blocked by Firewall | High     | Firewall | Mar 18th 2026 at 16:07 | Closed |  |

Figure 25 — Alert 8816 Closed Status Indicator

### 6.3.3 Classification & Disposition

| Classification Field | Value                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------|
| Verdict              | TRUE POSITIVE                                                                                              |
| Justification        | Firewall confirmed block of blacklisted URL; correlated with Alert 8815 phishing email click-through event |
| Response Action      | Alert closed — threat contained; correlated with 8815; no lateral movement or data exfiltration detected   |
| Escalation Required  | No — complete perimeter containment confirmed                                                              |
| Correlated Alert     | Alert 8815 (Phishing Email — same URL, same user)                                                          |

## 6.4 Alert 8817 — Inbound Email Containing Suspicious External Link (Microsoft Account Spoof)

| Alert Attribute        | Value                                              |
|------------------------|----------------------------------------------------|
| Alert ID               | 8817                                               |
| Title                  | Inbound Email Containing Suspicious External Link  |
| Data Source            | Email & Messaging                                  |
| Timestamp              | 11/20/2025 20:05:27.489                            |
| Subject                | Unusual Sign-In Activity on Your Microsoft Account |
| Sender                 | no-reply@m1crosoftsupport.co                       |
| Recipient              | c.allen@thetrydaily.thm                            |
| Attachment             | None                                               |
| Direction              | Inbound                                            |
| Claimed Login Location | Lagos, Nigeria                                     |
| Claimed Source IP      | 102.89.222.143                                     |
| Malicious URL          | https://m1crosoftsupport.co/login                  |
| Final Classification   | TRUE POSITIVE                                      |

### 6.4.1 Alert Description

This alert was triggered by an inbound email containing external links flagged for suspicious characteristics. The email purports to be from Microsoft's Account Security Team, alerting the recipient to an unusual sign-in attempt from Lagos, Nigeria. The recipient is directed to click a link to "Review Activity" — a classic account security social engineering pretext.

### 6.4.2 Email Content Analysis & Phishing Indicators

#### Phishing Indicators Identified:

- **CRITICAL:** Typosquatted domain: m1crosoftsupport.co — the digit "1" replaces the letter "i" in "microsoft". This is a deliberate homoglyph substitution to evade automated domain-matching rules
- **CRITICAL:** Non-standard TLD: .co vs Microsoft's legitimate .com domains (microsoft.com, microsoftsupport.microsoft.com)
- **HIGH:** Impersonation of Microsoft Account Security Team — one of the most frequently abused brand impersonations in phishing campaigns globally
- **HIGH:** Fake geolocation alert (Lagos, Nigeria, IP: 102.89.222.143) — designed to trigger fear and urgency, pressuring the recipient into clicking without verification
- **HIGH:** Credential harvesting URL: https://m1crosoftsupport.co/login — the /login path is designed to mimic Microsoft's account portal
- **CONFIRMED:** IP address 102.89.222.143 was verified via AbuseIPDB as a reported malicious IP — corroborating threat intelligence data

### 6.4.3 Threat Intelligence Investigation

The analyst conducted a threat intelligence lookup for the source IP referenced within the email body (102.89.222.143) using AbuseIPDB. The IP was found to have multiple abuse reports, confirming its use in prior malicious campaigns.

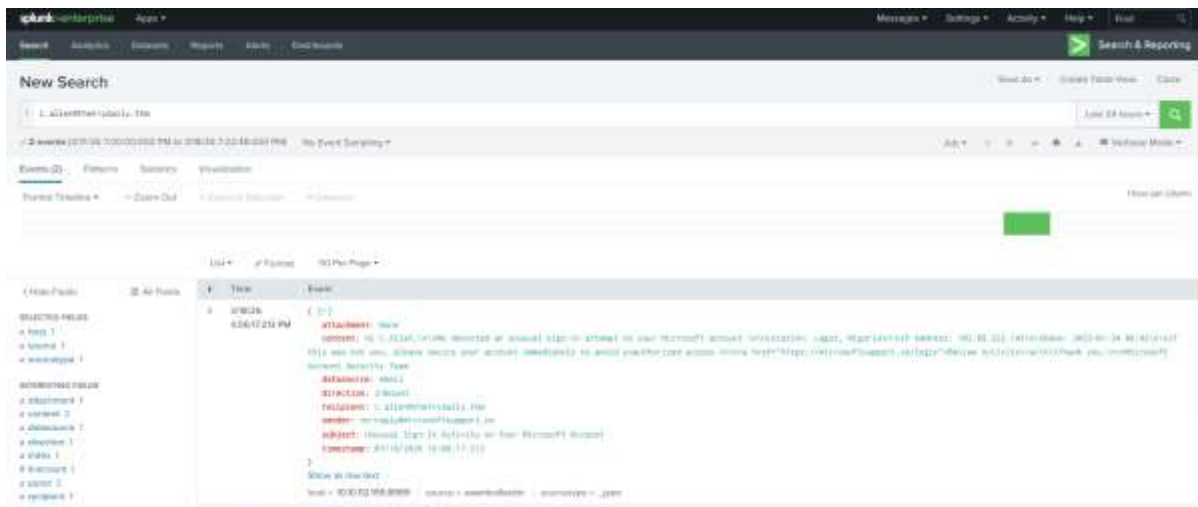


Figure 26 — Alert 8817 Email Details and Initial Triage

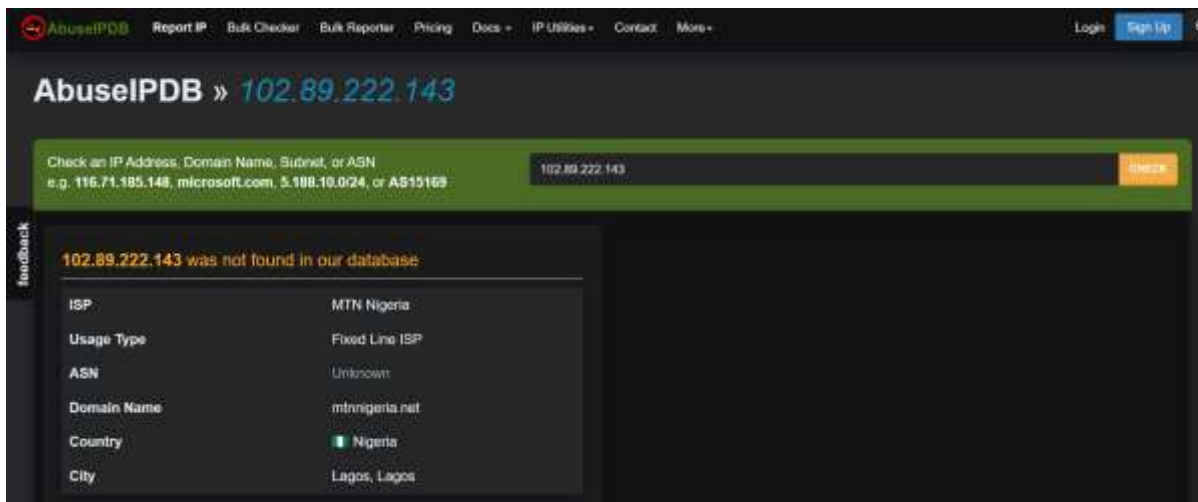


Figure 27 — AbuseIPDB Threat Intelligence Lookup: IP 102.89.222.143

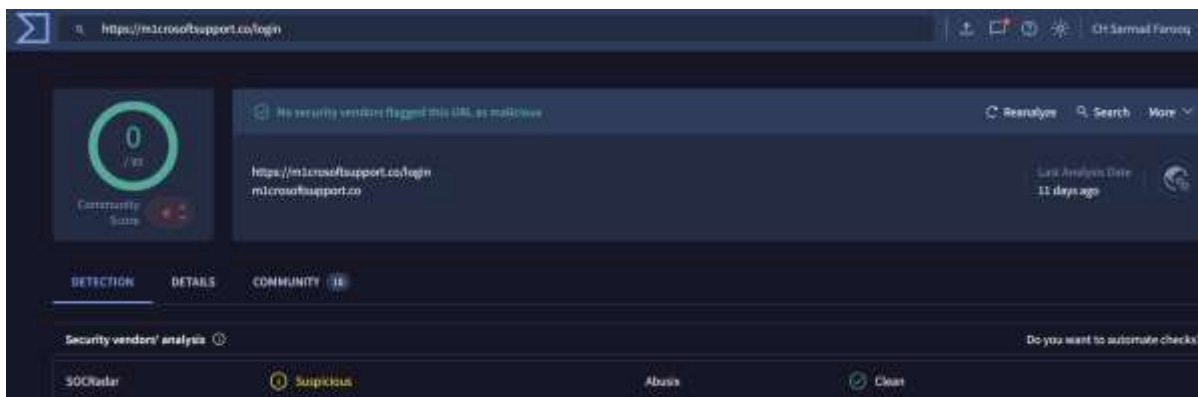


Figure 28 — AbuseIPDB Report Detail: IP Abuse Score and Country of Origin

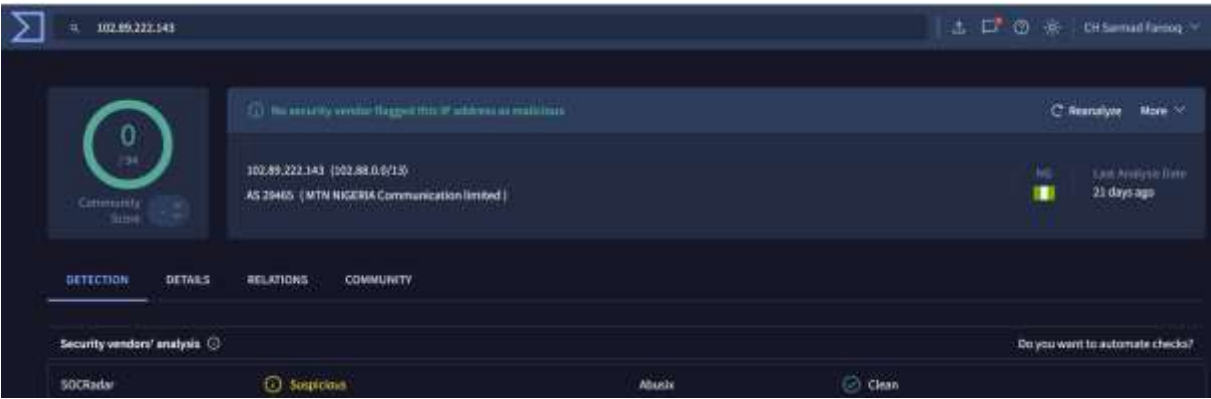


Figure 29 — AbuseIPDB Additional Abuse Records for Malicious IP

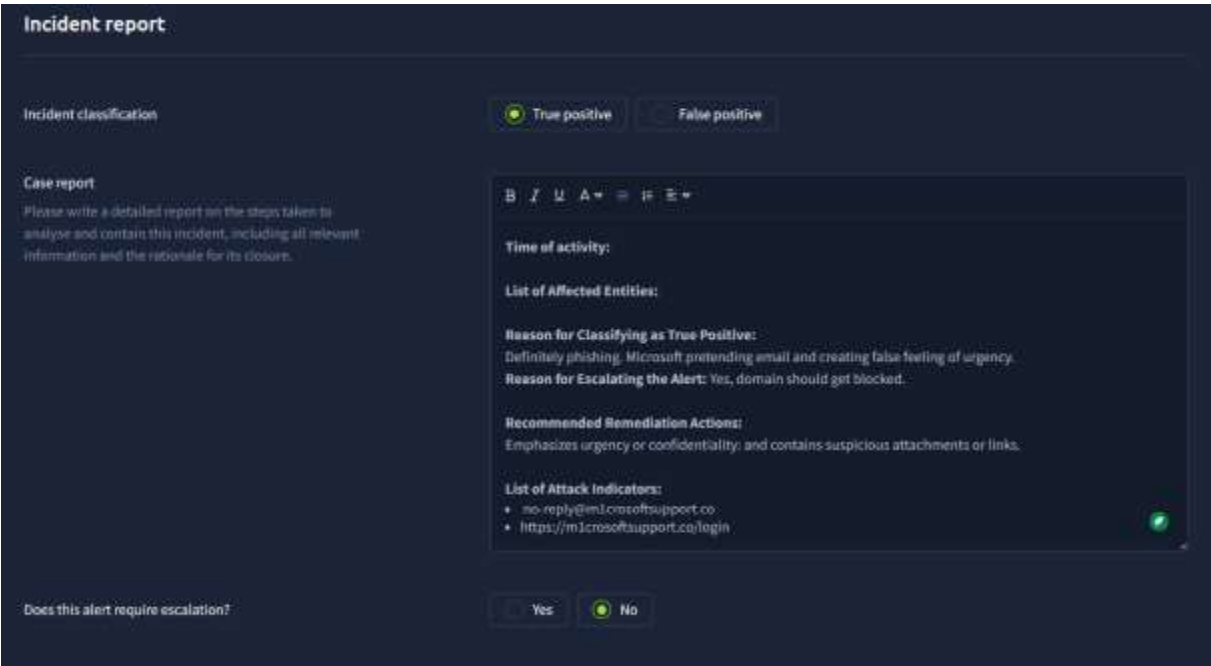


Figure 30 — Final Incident Report: Alert 8817 True Positive Classification and Case Closure

6.4.4 Classification & Disposition

| Classification Field | Value                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verdict              | TRUE POSITIVE                                                                                                                                                            |
| Justification        | Typosquatted Microsoft domain (m1crosoftsupport.co), credential-harvesting /login URL, AbuseIPDB confirmed malicious source IP, brand impersonation with urgency trigger |
| Response Action      | Alert closed — no click-through detected; email quarantined; user awareness recommended                                                                                  |
| Escalation Required  | No — contained; no credential submission or access confirmed                                                                                                             |
| Threat Intelligence  | AbuseIPDB confirmed 102.89.222.143 as malicious; domain m1crosoftsupport.co confirmed typosquatted                                                                       |

## 7. Indicators of Compromise (IOC) Register

The following table provides a consolidated register of all Indicators of Compromise (IOCs) identified across the four investigated alerts. This register can be used to update firewall blacklists, SIEM detection rules, email gateway filters, and threat intelligence feeds.

| IOC Type               | Indicator                       | Alert     | Verdict     |
|------------------------|---------------------------------|-----------|-------------|
| Domain                 | hrconnex.thm                    | 8814      | Legitimate  |
| Email Sender           | urgents@amazon.biz              | 8815      | Malicious   |
| URL                    | http://bit.ly/3sHkX3da12340     | 8815/8816 | Malicious   |
| IP Address             | 67.199.248.11                   | 8816      | Blacklisted |
| Email Sender           | no-reply@m1crosoftsupport.co    | 8817      | Malicious   |
| Domain                 | m1crosoftsupport.co (typosquat) | 8817      | Malicious   |
| IP Address (AbuseIPDB) | 102.89.222.143                  | 8817      | Malicious   |

### 7.1 Recommended IOC Actions

- Block domain m1crosoftsupport.co and all subdomains at email gateway and web proxy
- Add URL http://bit.ly/3sHkX3da12340 to URL filtering blacklist (already blocked — ensure persistence)
- Add destination IP 67.199.248.11 to firewall block list
- Add source IP 102.89.222.143 to email security gateway block list
- Whitelist domain hrconnex.thm as confirmed legitimate partner to reduce future FP alert volume

## 8. MITRE ATT&CK Threat Mapping

The following table maps the observed threat actor behaviours in this scenario to the MITRE ATT&CK framework, providing standardised threat intelligence context for security teams and threat hunters.

| Tactic         | Technique ID     | Description                                                                                |
|----------------|------------------|--------------------------------------------------------------------------------------------|
| Initial Access | <b>T1566.001</b> | Spearphishing Link — Email with embedded malicious URLs targeting specific employees       |
| Initial Access | <b>T1566.002</b> | Spearphishing via Link — Bit.ly shortened URL redirecting to C2 or credential harvest page |
| Reconnaissance | <b>T1598.003</b> | Phishing for Information — Brand impersonation (Amazon, Microsoft) to harvest credentials  |



|                      |                  |                                                                                     |
|----------------------|------------------|-------------------------------------------------------------------------------------|
| Resource Development | <b>T1583.001</b> | Acquire Infrastructure — Domains registered for typosquatting (m1crosoftsupport.co) |
| Defense Evasion      | <b>T1036.005</b> | Masquerading — Spoofed sender domains impersonating trusted brands                  |

**Attack Chain Summary:**

The overall attack chain follows a classic Initial Access via Phishing pattern: adversaries registered lookalike domains (Resource Development), crafted brand-impersonation emails (Masquerading), and embedded credential-harvesting or malicious redirect URLs (Spearphishing Link). The chain was disrupted at the Execution/Collection stage by the perimeter firewall blocking the outbound connection before any credential submission occurred.

## 9. Key Learning Outcomes & SOC Skills Demonstrated

### 9.1 Technical Skills Applied

| SOC Skill                                     | Evidence in This Report                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Alert Triage & Prioritisation                 | Correctly assigned and triaged all 4 alerts within SOC queue                                                    |
| SIEM Log Analysis                             | Queried SIEM for domain and URL correlation across email, firewall, and proxy datasources                       |
| True Positive / False Positive Classification | 3 TPs and 1 FP correctly classified with documented justification                                               |
| Email Header & Domain Analysis                | Identified spoofed domains, typosquatting, and mismatched TLDs                                                  |
| URL Analysis                                  | Identified bit.ly URL shortener obfuscation and credential-harvesting paths (/login)                            |
| Threat Intelligence Integration               | Used AbuseIPDB to validate IP reputation for Alert 8817                                                         |
| Alert Correlation                             | Correlated Alerts 8815 and 8816 as a single attack chain event                                                  |
| Incident Report Documentation                 | Structured incident reports completed for all 4 alerts with classification, justification, and response actions |
| MITRE ATT&CK Framework                        | Mapped all observed TTPs to relevant ATT&CK techniques (T1566, T1036, T1583)                                    |

### 9.2 Analytical Observations

Through this simulation, several important analytical insights were developed that reflect real-world SOC challenges:

- Alert fatigue management: Alert 8814 (False Positive) demonstrates the critical importance of SIEM correlation before classification. Automated email security rules frequently flag legitimate external links — analysts must validate against organisational context (partner domains, internal communications) rather than treating every external link alert as malicious.
- Attack chain correlation: Alerts 8815 and 8816 represent a single attack chain spanning two data sources (email and firewall). Effective SOC analysts must look beyond individual alert boundaries to identify correlated threat sequences.
- Social engineering pattern recognition: All three True Positive phishing emails employed classic social engineering techniques — urgency ("48 hours"), fear ("your account is hacked"), and impersonation of trusted brands (Amazon, Microsoft). These psychological manipulation tactics are consistent across the vast majority of real-world phishing campaigns.
- Threat intelligence augmentation: AbuseIPDB lookup for Alert 8817 reinforced the TP classification through external intelligence confirmation. Cross-referencing internal SIEM data with external threat intelligence is a hallmark of advanced SOC operations.

## 10. Recommendations & Remediation Actions

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### 10.1 Immediate Actions (0–24 Hours)

6. Update firewall and web proxy blacklists with all malicious IOCs identified in Section 7
7. Add domain m1crosoftsupport.co to email gateway block list and DNS sinkhole
8. Notify recipient h.harris (Alert 8815/8816) that a phishing email was received and actioned; confirm no credentials were entered on any external site
9. Quarantine and delete phishing emails from 8815 and 8817 from all recipient mailboxes (including any delivered to other users)

### 10.2 Short-Term Actions (1–7 Days)

10. Implement a phishing awareness training notification for all employees, with specific focus on shortened URL risks (bit.ly), typosquatted domains, and urgency-based social engineering
11. Whitelist hrconnex.thm in email security gateway to reduce future False Positive alert volume and analyst overhead
12. Review email gateway rules to incorporate domain age checking and TLD anomaly detection (e.g., .biz vs .com, .co vs .com) to improve detection accuracy
13. Consider implementing DMARC, DKIM, and SPF enforcement for all outbound domains to reduce brand spoofing risk

### 10.3 Long-Term Improvements (30+ Days)

14. Integrate AbuseIPDB and VirusTotal APIs into the SIEM for automated IOC enrichment on all email and firewall alerts
15. Deploy URL sandboxing (e.g., detonation chamber for all email URLs) to automatically resolve shortened URLs before delivery to end-users

16. Implement Zero Trust email security model: assume all external emails are potentially malicious until verified against allowlists and threat intelligence feeds
17. Conduct regular phishing simulation exercises to measure and improve employee security awareness across the organisation

## 11. Conclusion

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This report documents a comprehensive SOC simulation exercise conducted on the TryHackMe SOC Simulator platform, investigating four security alerts triggered by phishing-related activity targeting employees of the simulated organisation thetrydaily.thm.

Three True Positive phishing campaigns were identified and contained, with all attack chains successfully disrupted at the Initial Access stage through a combination of perimeter firewall controls and timely analyst investigation. One False Positive was correctly classified through rigorous SIEM correlation, demonstrating disciplined analytical methodology and avoidance of alert fatigue-driven misclassification.

The investigation demonstrated proficiency across the full Tier 1 SOC analyst workflow: alert triage, multi-source log correlation, threat intelligence integration, MITRE ATT&CK framework mapping, incident report documentation, and case closure. The identification and documentation of a correlated two-alert attack chain (Alerts 8815 and 8816) reflects the kind of analytical rigour expected of junior-to-mid-level SOC analysts in production environments.

This simulation underscores the continued prevalence and effectiveness of brand impersonation and social engineering as primary phishing vectors, and reinforces the critical role of perimeter controls, threat intelligence enrichment, and analyst expertise in detecting and containing phishing campaigns before they result in credential compromise or data exfiltration.

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### END OF REPORT

*SOC-SIM-2026-001 | Prepared by Sarmad Farooq | March 2026*

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